# Supplementary Material: *Measuring Local Acoustic Contribution in Speech LLMs via Word-Span Counterfactual Interventions*

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## S1. Additional experimental details

### A. Datasets, alignment, and target sampling

We use the same evaluation population and preprocessing as the main paper. An utterance is eligible after successful forced alignment when it contains at least six non-filler words and at least three non-target words remain after the sampled span is removed. One target span is sampled per eligible utterance. All three corpora use Montreal Forced Aligner word boundaries on a 10-ms grid. After excluding spoken fillers, the sampling-stratum probabilities for content, function, phrase (two to four words), number, boundary, and longest-duration targets are 0.30, 0.15, 0.25, 0.10, 0.10, and 0.10, respectively. The aligned span is replaced by silence or duration-matched MUSAN noise or music using 20-ms cosine crossfades. The same target spans and sampling seed are reused across replacement conditions; the primary LCCS and retention analyses use silence.

### B. Models and decoding

All systems are evaluated from their released checkpoints without additional training or fine-tuning. Decoding is greedy. Audio Flamingo Next and Voxtral Small receive the prompt "Transcribe the input speech, with no extra commentary."

### C. Target-category definitions

The benchmark records the sampling strata *function*, *content*, *number*, *boundary*, *longest-duration*, and *phrase*. Function words are selected from the benchmark's fixed function-word lexicon; the content stratum contains the remaining eligible non-number words. It is therefore a broad lexical class, not a part-of-speech annotation: nouns, verbs, adjectives, and adverbs are not separately labelled in the released evaluation metadata. Boundary and longest-duration are selection rules and can overlap linguistically with the function/content distinction, but each utterance belongs to exactly one sampled stratum.

TABLE S1
EVALUATION-SET COMPOSITION PER INTERVENTION FAMILY. EACH PHRASE SPANS TWO TO FOUR CONSECUTIVE REFERENCE WORDS; EVERY OTHER CATEGORY IS A SINGLE-WORD TARGET.

| Corpus | Spans/family | Single-word | Phrase |
|---|---|---|---|
| TED-LIUM | 1,053 | 762 (72.4%) | 291 (27.6%) |
| AMI | 5,181 | 3,811 (73.6%) | 1,370 (26.4%) |
| LibriSpeech | 2,693 | 2,000 (74.3%) | 693 (25.7%) |
| Total | 8,943 | 6,587 (73.7%) | 2,356 (26.3%) |

TABLE S2
SYSTEMS USED IN THE EVALUATION. PARAMETER COUNTS AND SYSTEM CLASSES MATCH THE MAIN PAPER.

| System | Parameters | Class |
|---|---|---|
| Whisper large-v3 | 1.5B | ASR |
| Qwen3-ASR | 1.7B | ASR |
| Granite Speech 4.1 | 2.3B | ASR |
| MERaLiON-3-ASR | 3.3B | ASR |
| Audio Flamingo Next | 8.3B | Multitask Speech LLM |
| Voxtral Small | 24.3B | Multitask Speech LLM |

## S2. ADDITIONAL RESULTS

### A. The full outcome–contribution matrix

The paper reports the $Y_i = C$ row of eq. (5) in the main paper. Tables S4 and S5 give all six cells, and Table S6 the contrast between the outcome rows. Applying the main-paper LCCS definition to the underlying unrounded $C$-row counts in Tables S4 and S5 reproduces every main-paper LCCS point estimate. Ratios formed from the displayed three-decimal cells can differ slightly only because the cells have already been rounded; bootstrap intervals remain in the main paper and are not duplicated here.

### B. Behaviour after evidence removal

Table S7 is the thresholded form of Fig. 3 and Table S8 gives the values plotted there. Tables S9 and S10 give the requested model-by-category analysis. Table S11 reports the broader all-target recovery diagnostic and is labelled separately because its population is not restricted to clean-correct targets.

### C. The intervention axis

Intervention type has little effect on the rank-based localization result but does affect the absolute prevalence of low-contribution correct outputs. Across silence, noise, and music replacement, $\mathrm{AUROC_span}$ changes only slightly (Table S12), indicating that AEC ranks edited spans relative to untouched words similarly across interventions. In contrast, LCCS at the fixed $\mathrm{AEF} = 2$threshold varies with both the replacement type and the model (Table S13). Silence removes the aligned speech evidence while introducing little competing acoustic content, whereas noise and music both remove the target evidence and insert new acoustic energy into the same interval. These replacements can produce a larger decrease in the reference-word likelihood, thereby increasing AEC and moving some targets above the fixed threshold. Consistent with this interpretation, silence produces the highest LCCS in 16 of the 18 model–corpus cells, although the magnitude of the effect differs across systems and is largest for MERaLiON-3-ASR. The lower LCCS observed with noise or music should therefore not be interpreted as evidence of stronger acoustic grounding. Rather, it shows that thresholded LCCS is conditional on the counterfactual replacement used to remove the original evidence. We consequently use silence as the primary intervention and treat noise and music as sensitivity analyses.

TABLE S3

THE OUTCOME–CONTRIBUTION MATRIX OF EQ. (5) IN THE MAIN PAPER, PER SYSTEM, FOR SINGLE-WORD TARGETS UNDER SILENCE REPLACEMENT. CELLS ARE $n_{ys}/N$ WITHIN EACH CORPUS AND ARE THEN AVERAGED OVER THE THREE CORPORA WITH EQUAL WEIGHT; ALL SIX CELLS IN A PANEL THEREFORE SUM TO ONE. THE $\Sigma$ COLUMN IS THE CLEAN FREE-DECODE OUTCOME DISTRIBUTION AND THE $\Sigma$ ROW IS THE OVERALL HIGH-CONTRIBUTION SHARE. PER-CORPUS $N$ IS 762/3,810–3,811/2,000 (TED-LIUM/AMI/LIBRISPEECH). $CU$ IS THE CORRECT-OUTPUT, LOW-CONTRIBUTION CELL THAT WER CANNOT DISTINGUISH FROM $CG$. LCCS (%) IS THE MEAN OF THE THREE PER-CORPUS $CU/(CG{+}CU)$ RATIOS; BECAUSE AVERAGING AND DIVISION DO NOT COMMUTE, IT DIFFERS FROM THE RATIO OF THE DISPLAYED AVERAGED CELLS BY AT MOST 0.19 POINTS.

| **Audio Flamingo Next** | | | LCCS 33.1 |
|---|---|---|---|
| | $G$ | $U$ | $\Sigma$ |
| $C$ correct | 0.629 | 0.311 | 0.941 |
| $E$ substituted | 0.034 | 0.012 | 0.047 |
| $D$ deleted | 0.007 | 0.006 | 0.013 |
| $\Sigma$ | 0.671 | 0.329 | 1.000 |

| **Qwen3-ASR** | | | LCCS 33.8 |
|---|---|---|---|
| | $G$ | $U$ | $\Sigma$ |
| $C$ correct | 0.621 | 0.318 | 0.938 |
| $E$ substituted | 0.032 | 0.010 | 0.041 |
| $D$ deleted | 0.014 | 0.007 | 0.020 |
| $\Sigma$ | 0.666 | 0.334 | 1.000 |

| **Whisper large-v3** | | | LCCS 36.4 |
|---|---|---|---|
| | $G$ | $U$ | $\Sigma$ |
| $C$ correct | 0.586 | 0.335 | 0.921 |
| $E$ substituted | 0.034 | 0.011 | 0.045 |
| $D$ deleted | 0.019 | 0.015 | 0.035 |
| $\Sigma$ | 0.639 | 0.361 | 1.000 |

| **Voxtral Small** | | | LCCS 37.5 |
|---|---|---|---|
| | $G$ | $U$ | $\Sigma$ |
| $C$ correct | 0.576 | 0.346 | 0.922 |
| $E$ substituted | 0.033 | 0.012 | 0.045 |
| $D$ deleted | 0.019 | 0.013 | 0.032 |
| $\Sigma$ | 0.629 | 0.371 | 1.000 |

| **Granite Speech 4.1** | | | LCCS 43.7 |
|---|---|---|---|
| | $G$ | $U$ | $\Sigma$ |
| $C$ correct | 0.536 | 0.417 | 0.952 |
| $E$ substituted | 0.025 | 0.011 | 0.036 |
| $D$ deleted | 0.005 | 0.006 | 0.012 |
| $\Sigma$ | 0.566 | 0.434 | 1.000 |

| **MERaLiON-3-ASR** | | | LCCS 48.7 |
|---|---|---|---|
| | $G$ | $U$ | $\Sigma$ |
| $C$ correct | 0.474 | 0.454 | 0.929 |
| $E$ substituted | 0.038 | 0.017 | 0.054 |
| $D$ deleted | 0.008 | 0.008 | 0.017 |
| $\Sigma$ | 0.521 | 0.479 | 1.000 |

TABLE S4

OUTCOME–CONTRIBUTION MATRIX FOR SINGLE-WORD TARGETS UNDER SILENCE REPLACEMENT. VALUES ARE $n_{ys}/N$; THE SIX CELLS IN EACH CORPUS–SYSTEM ROW SUM TO ONE BEFORE ROUNDING. $Y_i \in \{C, E, D\}$ IS THE CLEAN FREE-DECODE OUTCOME (CORRECT, SUBSTITUTED, OR DELETED), AND $G/U$ DENOTES HIGH/LOW CONTRIBUTION AT $\mathrm{AEC}_i > \ln 2$ (EQUIVALENTLY $\mathrm{AEF}_i > 2$). INSERTIONS HAVE NO REFERENCE TARGET SPAN AND ARE NOT REPRESENTED.

| System | $n_{CG}/N$ | $n_{CU}/N$ | $n_{EG}/N$ | $n_{EU}/N$ | $n_{DG}/N$ | $n_{DU}/N$ | $N$ |
|---|---|---|---|---|---|---|---|
| *TED-LIUM* | | | | | | | |
| Audio Flamingo Next | 0.646 | 0.318 | 0.013 | 0.007 | 0.008 | 0.009 | 762 |
| Qwen3-ASR | 0.583 | 0.374 | 0.017 | 0.007 | 0.012 | 0.008 | 762 |
| Whisper large-v3 | 0.602 | 0.346 | 0.016 | 0.007 | 0.018 | 0.010 | 762 |
| Voxtral Small | 0.617 | 0.331 | 0.018 | 0.007 | 0.020 | 0.008 | 762 |
| Granite Speech 4.1 | 0.504 | 0.453 | 0.014 | 0.008 | 0.009 | 0.012 | 762 |
| MERaLiON-3-ASR | 0.413 | 0.551 | 0.017 | 0.009 | 0.000 | 0.009 | 762 |
| *AMI* | | | | | | | |
| Audio Flamingo Next | 0.599 | 0.307 | 0.055 | 0.021 | 0.011 | 0.007 | 3,811 |
| Qwen3-ASR | 0.641 | 0.261 | 0.046 | 0.015 | 0.025 | 0.012 | 3,811 |
| Whisper large-v3 | 0.540 | 0.325 | 0.050 | 0.018 | 0.035 | 0.031 | 3,811 |
| Voxtral Small | 0.551 | 0.319 | 0.048 | 0.020 | 0.034 | 0.028 | 3,811 |
| Granite Speech 4.1 | 0.532 | 0.401 | 0.034 | 0.020 | 0.006 | 0.007 | 3,810 |
| MERaLiON-3-ASR | 0.497 | 0.380 | 0.059 | 0.029 | 0.022 | 0.013 | 3,811 |
| *LibriSpeech* | | | | | | | |
| Audio Flamingo Next | 0.642 | 0.310 | 0.035 | 0.009 | 0.003 | 0.001 | 2,000 |
| Qwen3-ASR | 0.638 | 0.319 | 0.032 | 0.007 | 0.004 | 0.001 | 2,000 |
| Whisper large-v3 | 0.615 | 0.334 | 0.036 | 0.007 | 0.004 | 0.004 | 2,000 |
| Voxtral Small | 0.560 | 0.389 | 0.033 | 0.011 | 0.005 | 0.003 | 2,000 |
| Granite Speech 4.1 | 0.572 | 0.396 | 0.025 | 0.006 | 0.001 | 0.000 | 2,000 |
| MERaLiON-3-ASR | 0.513 | 0.431 | 0.038 | 0.011 | 0.004 | 0.003 | 2,000 |

TABLE S5

OUTCOME–CONTRIBUTION MATRIX FOR SPAN-MACRO TARGETS UNDER SILENCE REPLACEMENT. VALUES ARE $n_{ys}/N$; THE SIX CELLS IN EACH CORPUS–SYSTEM ROW SUM TO ONE BEFORE ROUNDING. EACH SAMPLED SPAN CONTRIBUTES ONE OBSERVATION. A PHRASE IS CORRECT ONLY WHEN ALL OF ITS REFERENCE WORDS ARE CORRECT, AND ITS AEC IS AVERAGED OVER THOSE WORDS. $G/U$ DENOTES HIGH/LOW CONTRIBUTION AT $\mathrm{AEC}_i > \ln 2$.

| System | $n_{CG}/N$ | $n_{CU}/N$ | $n_{EG}/N$ | $n_{EU}/N$ | $n_{DG}/N$ | $n_{DU}/N$ | $N$ |
|---|---|---|---|---|---|---|---|
| *TED-LIUM* | | | | | | | |
| Audio Flamingo Next | 0.713 | 0.242 | 0.028 | 0.005 | 0.006 | 0.007 | 1,053 |
| Qwen3-ASR | 0.666 | 0.283 | 0.032 | 0.005 | 0.009 | 0.006 | 1,053 |
| Whisper large-v3 | 0.668 | 0.270 | 0.035 | 0.007 | 0.013 | 0.008 | 1,053 |
| Voxtral Small | 0.677 | 0.259 | 0.035 | 0.009 | 0.014 | 0.006 | 1,053 |
| Granite Speech 4.1 | 0.588 | 0.354 | 0.034 | 0.009 | 0.007 | 0.009 | 1,053 |
| MERaLiON-3-ASR | 0.518 | 0.443 | 0.024 | 0.009 | 0.000 | 0.007 | 1,053 |
| *AMI* | | | | | | | |
| Audio Flamingo Next | 0.645 | 0.235 | 0.087 | 0.018 | 0.010 | 0.005 | 5,181 |
| Qwen3-ASR | 0.668 | 0.198 | 0.091 | 0.014 | 0.020 | 0.009 | 5,181 |
| Whisper large-v3 | 0.568 | 0.252 | 0.108 | 0.019 | 0.030 | 0.024 | 5,181 |
| Voxtral Small | 0.578 | 0.248 | 0.105 | 0.019 | 0.028 | 0.021 | 5,181 |
| Granite Speech 4.1 | 0.597 | 0.315 | 0.060 | 0.017 | 0.006 | 0.006 | 5,180 |
| MERaLiON-3-ASR | 0.542 | 0.296 | 0.108 | 0.027 | 0.017 | 0.010 | 5,180 |
| *LibriSpeech* | | | | | | | |
| Audio Flamingo Next | 0.695 | 0.245 | 0.048 | 0.009 | 0.003 | 0.000 | 2,693 |
| Qwen3-ASR | 0.690 | 0.250 | 0.051 | 0.006 | 0.003 | 0.000 | 2,693 |
| Whisper large-v3 | 0.660 | 0.267 | 0.059 | 0.007 | 0.004 | 0.003 | 2,693 |
| Voxtral Small | 0.615 | 0.316 | 0.053 | 0.010 | 0.004 | 0.002 | 2,693 |
| Granite Speech 4.1 | 0.638 | 0.317 | 0.038 | 0.006 | 0.001 | 0.000 | 2,693 |
| MERaLiON-3-ASR | 0.578 | 0.356 | 0.052 | 0.010 | 0.003 | 0.002 | 2,693 |

TABLE S6

HIGH-CONTRIBUTION RATE BY CLEAN-DECODE OUTCOME, POOLED OVER THE 18 SYSTEM×CORPUS CELLS UNDER SILENCE REPLACEMENT. BRACKETS ARE 95 % WILSON INTERVALS THAT TREAT THE POOLED TARGET ROWS AS OBSERVATIONS; BECAUSE THEY DO NOT ACCOUNT FOR MODEL, CORPUS, OR UTTERANCE CLUSTERING, THEY ARE DESCRIPTIVE RATHER THAN INFERENTIAL. SUBSTITUTED TARGETS HAVE A HIGHER REFERENCE-WORD CONTRIBUTION RATE THAN CORRECT TARGETS IN ALL 18 SPAN-MACRO CELLS AND 17 OF 18 SINGLE-WORD CELLS. THIS SHOWS THAT MANY SUBSTITUTIONS OCCUR DESPITE EVIDENCE FOR THE REFERENCE WORD UNDER TEACHER FORCING; FREE DECODING NEVERTHELESS SELECTS ANOTHER WORD.

| | Span-macro | | Single-word | |
|---|---|---|---|---|
| Clean outcome $Y_i$ | $N$ | $\Pr(g_i{=}1 \mid Y_i)$ | $N$ | $\Pr(g_i{=}1 \mid Y_i)$ |
| Correct $(C)$ | 47,774 | 69.3 [68.9, 69.7] | 36,209 | 62.0 [61.5, 62.5] |
| Substituted $(E)$ | 4,665 | 83.7 [82.7, 84.8] | 2,190 | 72.3 [70.4, 74.2] |
| Deleted $(D)$ | 1,121 | 59.2 [56.4, 62.1] | 1,038 | 57.7 [54.7, 60.7] |

TABLE S7

RETENTION AFTER EVIDENCE REMOVAL AT THE FIXED THRESHOLD $\mathrm{AEF} > 2$, BY SYSTEM (T/A/L: TED-LIUM / AMI / LIBRISPEECH). $\Pr(R_i{=}1 \mid g_i{=}0)$ AND $\Pr(R_i{=}1 \mid g_i{=}1)$ OVER CLEAN-CORRECT TARGETS, SPAN-MACRO, SILENCE INTERVENTION. THE GAP IS THE DIFFERENCE IN POINTS. THIS IS THE THRESHOLDED FORM OF FIG. 3, WHICH BINS THE SAME TARGETS BY AEC QUINTILE WITH NO THRESHOLD IN THE ESTIMATOR.

| System | low-AEC retained | high-AEC retained | gap |
|---|---|---|---|
| Audio Flamingo Next | 90.6/69.3/82.0 | 4.9/3.2/6.8 | 85.7/66.0/75.2 |
| Qwen3-ASR | 91.3/85.7/87.2 | 4.4/2.5/4.7 | 86.9/83.1/82.5 |
| Whisper large-v3 | 96.1/85.1/87.5 | 17.1/7.7/14.0 | 79.1/77.5/73.5 |
| Voxtral Small | 95.2/87.8/84.7 | 17.3/7.5/9.3 | 78.0/80.3/75.4 |
| Granite Speech 4.1 | 94.1/80.5/87.0 | 2.9/3.5/9.4 | 91.2/77.0/77.7 |
| MERaLiON-3-ASR | 92.3/85.6/84.2 | 8.1/5.2/13.5 | 84.2/80.4/70.7 |
| *over the six* | *93.3/82.3/85.4* | *9.1/4.9/9.6* | *84.2/77.4/75.8* |
| *range, 18 cells* | *69.3–96.1* | *2.5–17.3* | |

TABLE S8

RETENTION BY AEC QUINTILE, MACRO-AVERAGED OVER THE SIX SYSTEMS (CLEAN-CORRECT TARGETS, SILENCE INTERVENTION) — THE POINT ESTIMATES BEHIND FIG. 3 IN THE MAIN PAPER. QUINTILES ARE FORMED WITHIN EACH SYSTEM×CORPUS CELL, SO $Q_1$ IS THAT CELL'S OWN LOWEST-AEC FIFTH. THE MAIN PAPER RETAINS THE UNCERTAINTY DISPLAY; THIS TABLE INTENTIONALLY REPORTS ONLY THE MATCHING POINT ESTIMATES.

| Corpus | $Q_1$ | $Q_2$ | $Q_3$ | $Q_4$ | $Q_5$ |
|---|---|---|---|---|---|
| TED-LIUM | 83.2 | 60.4 | 19.2 | 3.2 | 0.8 |
| AMI | 75.5 | 36.2 | 6.2 | 2.8 | 1.2 |
| LibriSpeech | 82.9 | 48.7 | 15.0 | 4.0 | 1.3 |

TABLE S9

LCCS (%) BY SYSTEM AND SAMPLED SINGLE-WORD CATEGORY UNDER SILENCE REPLACEMENT. EACH CELL IS FIRST COMPUTED WITHIN EACH CORPUS AS $\Pr(\mathrm{AEC}_i \le \ln 2 \mid Y_i = C, \text{CATEGORY})$ AND THEN MACRO-AVERAGED OVER TED-LIUM, AMI, AND LIBRISPEECH. FUNCTION WORDS HAVE THE HIGHEST LCCS FOR EVERY SYSTEM; LONGEST-DURATION TARGETS HAVE THE LOWEST. "CONTENT" IS THE BENCHMARK'S BROAD NON-FUNCTION STRATUM, NOT A NOUN-ONLY PART-OF-SPEECH CLASS.

| System | Function | Content | Number | Boundary | Longest |
|---|---|---|---|---|---|
| Audio Flamingo Next | 65.2 | 24.4 | 40.6 | 33.3 | 6.9 |
| Qwen3-ASR | 67.9 | 25.8 | 50.4 | 27.1 | 7.4 |
| Whisper large-v3 | 68.3 | 28.9 | 49.1 | 30.4 | 13.5 |
| Voxtral Small | 67.9 | 31.4 | 51.9 | 28.5 | 15.2 |
| Granite Speech 4.1 | 76.0 | 34.7 | 61.5 | 43.7 | 17.3 |
| MERaLiON-3-ASR | 76.9 | 43.3 | 64.6 | 44.7 | 21.1 |

TABLE S10

RETENTION (%) AFTER SILENCE REPLACEMENT BY SYSTEM AND SAMPLED SINGLE-WORD CATEGORY. THE POPULATION IS RESTRICTED TO CLEAN-CORRECT TARGETS, AND A TARGET IS RETAINED ONLY WHEN THE EDITED-AUDIO FREE DECODE RE-EMITS IT AT ITS ALIGNED REFERENCE POSITION. EACH CELL IS MACRO-AVERAGED OVER THE THREE CORPUS-SPECIFIC RATES. THIS USES THE SAME RETENTION DEFINITION AND POPULATION AS THE MAIN PAPER, UNLIKE AN UNCONDITIONAL RECOVERY RATE OVER ALL TARGETS.

| System | Function | Content | Number | Boundary | Longest |
|---|---|---|---|---|---|
| Audio Flamingo Next | 66.1 | 24.5 | 43.8 | 23.6 | 6.4 |
| Qwen3-ASR | 68.4 | 26.6 | 50.4 | 24.8 | 7.2 |
| Whisper large-v3 | 76.3 | 38.0 | 61.3 | 37.5 | 15.3 |
| Voxtral Small | 76.4 | 38.3 | 60.3 | 34.8 | 14.6 |
| Granite Speech 4.1 | 77.2 | 36.0 | 63.8 | 36.8 | 13.9 |
| MERaLiON-3-ASR | 78.0 | 46.5 | 64.3 | 42.8 | 22.9 |

TABLE S11

ANCHORED REFERENCE-WORD RECOVERY (%) OVER ALL SAMPLED TARGETS BY THE DURATION OF THE REMOVED SPAN, MACRO-AVERAGED OVER THE SIX SYSTEMS UNDER SILENCE REPLACEMENT. THIS DESCRIPTIVE DIAGNOSTIC IS NOT THE CLEAN-CORRECT RETENTION ESTIMAND IN TABLES S7 AND S10. BINS ARE CORPUS-SPECIFIC DURATION QUARTILES, WITH THEIR EDGES SHOWN BELOW EACH ROW.

| Corpus | Q1 (shortest) | Q2 | Q3 | Q4 (longest) |
|---|---|---|---|---|
| TED-LIUM | 79.3 | 54.1 | 29.8 | 22.6 |
| *bin* | *<0.22 s* | *0.22–0.42 s* | *0.42–0.66 s* | *≥0.66 s* |
| AMI | 60.6 | 36.8 | 26.7 | 16.9 |
| *bin* | *<0.20 s* | *0.20–0.34 s* | *0.34–0.55 s* | *≥0.55 s* |
| LibriSpeech | 68.8 | 30.1 | 14.6 | 8.6 |
| *bin* | *<0.25 s* | *0.25–0.43 s* | *0.43–0.64 s* | *≥0.64 s* |

TABLE S12

$\mathrm{AUROC}_{\mathrm{span}}$ BY INTERVENTION, MACRO-AVERAGED OVER THE SIX SYSTEMS (SINGLE-WORD TARGETS, WITHIN-UTTERANCE ESTIMAND). AVERAGING THE UNDERLYING UNROUNDED ENTRIES ACROSS INTERVENTIONS YIELDS THE T/A/L CORPUS TRIPLET IN THE MAIN-PAPER AUROC TABLE; AVERAGING ACROSS CORPORA YIELDS ITS S/N/M INTERVENTION TRIPLET. THE SENSITIVITY OF EVERY SCORE IS ESSENTIALLY UNCHANGED BY THE FILLER — AEC MOVES BY AT MOST 0.017 ACROSS THE THREE — WHICH LICENSES POOLING THE INTERVENTIONS FOR THE LOCALIZATION ANALYSIS IN THE MAIN PAPER. THIS DOES NOT LICENSE POOLING THEM FOR LCCS, WHICH IS A THRESHOLD-CROSSING PREVALENCE RATHER THAN A RANKING; SEE TABLE S13.

| Corpus | Score | Silence | Noise | Music |
|---|---|---|---|---|
| TED-LIUM | Prior surprisal | .58 | .58 | .58 |
| | Entropy $H$ | .88 | .89 | .89 |
| | $\Delta H$ | .91 | .91 | .91 |
| | AEC | .92 | .94 | .94 |
| AMI | Prior surprisal | .55 | .55 | .55 |
| | Entropy $H$ | .81 | .82 | .83 |
| | $\Delta H$ | .85 | .86 | .86 |
| | AEC | .89 | .90 | .89 |
| LibriSpeech | Prior surprisal | .54 | .54 | .54 |
| | Entropy $H$ | .87 | .89 | .89 |
| | $\Delta H$ | .93 | .94 | .94 |
| | AEC | .89 | .90 | .90 |

TABLE S13

LCCS (%) BY INTERVENTION FOR SINGLE-WORD TARGETS PRESENT IN ALL THREE INTERVENTIONS (T/A/L: TED-LIUM / AMI / LIBRISPEECH). SILENCE YIELDS THE HIGHEST LCCS IN 16 OF THE 18 SYSTEM×CORPUS CELLS, BECAUSE IT REMOVES ACOUSTIC EVIDENCE WITHOUT SUBSTITUTING COMPETING EVIDENCE. THE SYSTEM SPREAD IS WIDEST UNDER SILENCE AND IS DRIVEN MAINLY BY MERALION-3-ASR. SPAN-MACRO TARGETS GIVE THE SAME QUALITATIVE PATTERN AT LOWER ABSOLUTE LEVELS.

| System | Silence | Noise | Music |
|---|---|---|---|
| Audio Flamingo Next | 33.0/33.9/32.5 | 29.0/32.6/29.4 | 31.6/34.8/32.4 |
| Qwen3-ASR | 39.1/28.9/33.3 | 34.8/29.4/31.2 | 34.4/28.6/30.8 |
| Whisper large-v3 | 36.5/37.5/35.2 | 30.3/30.8/27.4 | 29.5/30.2/26.7 |
| Voxtral Small | 34.9/36.7/40.9 | 29.2/31.5/32.6 | 29.5/32.4/33.1 |
| Granite Speech 4.1 | 47.3/43.0/40.9 | 38.4/38.1/35.6 | 38.5/38.3/35.6 |
| MERaLiON-3-ASR | 57.1/43.4/45.7 | 31.8/30.0/32.6 | 32.7/29.9/33.4 |
| *over the six* | *41.3/37.2/38.1* | *32.3/32.1/31.5* | *32.7/32.4/32.0* |
| *spread over the six* | *24.2/14.4/13.1* | *9.4/8.7/8.2* | *9.1/9.7/8.8* |
| *band, 18 cells* | *28.9–57.1* | *27.4–38.4* | *26.7–38.5* |